THAR INSTITUTE OF ENGINEERING, SCIENCES & TECHNOLOGY
Constituent College of NED University of Engineering & Technology

SOFTWARE REQUIREMENTS SPECIFICATION

Bloodbound – Blood Donor & Camp Coordination System

Course Title:	Software Engineering
Course Code:	CT-365
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1. System Roles & Overview

Bloodbound streamlines emergency donor matching and camp management. The platform identifies four key roles:

- **Donor:** Registers, maintains profile/health status, receives proximity alerts, accepts emergency dispatches, and pre-books camp slots.
- **Hospital Staff / Patient Requester:** Creates urgent requests (specifying blood group, units, component), tracks donor responses in real-time.
- **Camp Organizer:** Schedules drives, manages slot capacities, and checks in donors.
- **Admin:** Moderates accounts, manages system settings, and inspects audit logs.

2. Functional Requirements (FR)

ID	Module	Description	Priority
FR-01	Auth	User registration, login (JWT + bcrypt), and role assignment.	High
FR-02	Profile	Donor profile management (blood group, location, last donation date).	High
FR-03	Request	Urgent request posting by hospital staff / patient requesters.	High
FR-04	Matching	Geospatial matching engine triggers alerts within configurable radius.	High
FR-05	Alerts	Broadcast notifications to eligible donors upon request creation.	High
FR-06	Response	Donor accepts request; system issues digital transit pass.	High
FR-07	Camps	Schedule donation drives with date, location, and total capacity.	Medium
FR-08	Slots	Donor slot pre-booking with real-time capacity checks.	Medium

3. User Stories (SCRUM-18)

- **US-01 (Donor Profile):** As a registered donor, I want to update my profile and donation date so that I receive accurate match alerts.
- **US-02 (Emergency Request):** As hospital staff or a patient relative, I want to submit blood requests so nearby donors are alerted.
- **US-03 (Camp Slot):** As a donor, I want to pre-book a camp time slot to avoid long waiting queues.
- **US-04 (Admin Audit):** As an admin, I want to view system activity logs to ensure platform integrity.

4. Requirements Traceability Matrix (SCRUM-19)

Requirement ID	Requirement Description	User Story	System Module
FR-01	User Auth & Role Access	US-01, US-04	`/api/auth` (JWT + bcrypt)
FR-02	Donor Profile Management	US-01	`/api/donors/profile`
FR-03	Emergency Request Submission	US-02	`/api/requests/create`
FR-04	Geospatial Donor Matching Engine	US-02	`MatchingEngine.js`
FR-05	Broadcast Notifications	US-02	`/api/alerts/broadcast`
FR-06	Donor Response & Transit Pass	US-02	`/api/requests/accept`
FR-07 / 08	Camp Drives & Slot Pre-Booking	US-03	`/api/camps/book`

5. Non-Functional Requirements (NFR)

ID	Category	Specification	Target Criterion
NFR-01	Performance	Emergency alert dispatch latency	Broadcast dispatched in < 3s
NFR-02	Security	Data encryption at rest and in transit	AES-256 & HTTPS / TLS 1.3
NFR-03	Uptime	Core system availability during emergency hours	99.9% operational availability
NFR-04	Scalability	Concurrent response handling during alerts	Handles 500+ concurrent connections

6. System Activity Diagram (Emergency Flow)



7. Sequence Diagram (Emergency Matching)

